

AJAY TATA

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Summary

Master's student in Data Science with strong SQL fundamentals and experience building end-to-end data pipelines using Python and pandas. Skilled in cleaning and transforming large raw datasets, standardizing schemas, and creating reusable SQL views to support data warehousing. Developed portfolio projects-including crime data analysis and NLP pipelines-demonstrating proficiency with Git and a keen ability to translate complex data into structured intelligence assets.

Education

University of colorado boulder

Aug 2024 - May 2026

Masters, Data Science

Boulder, co

- **GPA:** 3.8/4

- **Coursework:** Data Visualization, Machine Learning, Statistical Analysis, Data Mining, Neural Nets and Deep Learning, Data Structures

Gokaraju Rangaraju Institute of Engineering and Technology

Aug 2021 - May 2024

Computer science Engineering, Computer science Engineering

Hyderabad, India

- **GPA:** 8.4/10

- **Coursework:** Data Visualization, Machine Learning, Statistical Analysis, Data Mining, Neural Nets and Deep Learning, Data Structures

Projects

PsyCoMark Transformer-Based Text Classification Project

Aug 2025 - Jan 2025

- Built an end-to-end data processing and machine learning pipeline using Python, staging transformed text data in BigQuery and applying data warehousing concepts for scalable storage.
- Cleaned and transformed raw text data using pandas in Python, performing feature engineering and tokenization to support robust model training.
- Developed and trained machine learning (scikit-learn) and deep learning (TensorFlow) models for text classification and information extraction, applying statistical analysis to optimize performance.
- Evaluated model performance and optimized workflows to ensure scalable, reproducible, production-ready pipelines managed with Git version control.

Mitigating Bias in NLP Models for Depression Prediction

May 2025

- Analyzed patient interview transcripts using NLP techniques (TF-IDF, VADER sentiment analysis) to extract linguistic and semantic features.
- Developed and evaluated Random Forest, Neural Network, and GPT-2 models using scikit-learn and TensorFlow, assessing predictive accuracy for depression severity and demographic classification.
- Achieved 90%+ accuracy in gender classification and strong correlation in depression severity estimation.
- Identified and mitigated bias across gender and race groups, improving fairness in model outcomes.
- Cleaned, merged, and processed large-scale E-DAIC clinical dataset using SQL queries in BigQuery and pandas, ensuring data integrity and enabling reproducible ETL workflows under Git version control.

Los Angeles Crime Data Analysis

Aug 2024 - Jan 2025

- Analyzed over 800K crime records from the Los Angeles Open Data portal to explore trends across time, location, and type.
- Performed data cleaning and exploratory data analysis (EDA) using Pandas and NumPy to prepare the dataset for insights.
- Built interactive dashboards and data visualizations using Plotly and Seaborn to showcase crime patterns and distributions.
- Identified high-crime zones, peak hours, and common offenses to inform potential strategies for public safety planning.
- Documented the project through a structured Google Site to communicate findings and methodology in a portfolio-friendly format.

Technical Skills

- **Programming/Query Language:** Python, Numpy, Pandas, BeautifulSoup, matplotlib, scikit-learn, Request, SQL, JAVA, C, R
- **Machine Learning:** Regression, classification, clustering, Modevaluation, Decision tree
- **Data Visualization:** Matplotlib, Seaborn, Plotly, ggplot2, Tableau, Microsoft Power BI
- **Data Analysis:** Exploratory Data Analysis(EDA), Feature extraction, Data Cleaning, Handling Outlier, Excel
- **Soft Skills:** Problem solving, Data Storytelling, Leadership, Time Management
- **Data Engineering & Platforms:** Data Warehousing, Git, Google BigQuery, Dbt, API

Publications

- A Spatial-Attentional Generative Adversarial Network for Building Structure Inference. *Co-authored a research paper proposing a GAN model with spatial attention to infer building layouts from UAV imagery, achieving 0.83 IoU with reduced structural deviation. Introduced a hybrid loss function (Dice + Hough-transform penalty) to improve straight-line feature completeness in architectural inference. Publication, AIP 2024..*